

Name :- Harsh Chandrakant Borhade

Roll No :- 25092005

Bus Ticket Counter Simulation using Queue

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX 5
```

```
char queue[MAX][50];
```

```
int front = -1, rear = -1;
```

```
int isFull() {
```

```
    return (rear == MAX - 1);
```

```
}
```

```
int isEmpty() {
```

```
    return (front == -1 || front > rear);
```

```
}
```

```
void enqueue(char name[]) {
```

```
    if (isFull()) {
```

```
    printf("Queue is full! No more passengers can join.\n");  
    return;  
}  
if (front == -1)  
    front = 0;  
strcpy(queue[++rear], name);  
printf("Passenger %s joined the queue.\n", name);  
}
```

```
void dequeue() {  
    if (isEmpty()) {  
        printf("No passengers to serve!\n");  
        return;  
    }  
    printf("Passenger %s got the ticket and left the  
queue.\n", queue[front++]);  
}
```

```
void display() {  
    if (isEmpty()) {  
        printf("No passengers waiting in line.\n");  
        return;  
    }  
}
```

```

printf("\n Passengers in Queue (Front to Rear):\n");
for (int i = front; i <= rear; i++)
    printf(" %d. %s\n", i - front + 1, queue[i]);
printf("-----\n");
}

int main() {
    int choice;
    char name[50];
    printf(" Bus Ticket Counter Simulation using Queue \n");

    while (1) {
        printf("\n1. New Passenger Arrives\n2. Serve
Passenger\n3. Show Queue\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar(); // clear newline

        switch (choice) {
            case 1:
                printf("Enter passenger name: ");
                gets(name);
                enqueue(name);

```

```
        break;
    case 2:
        dequeue();
        break;
    case 3:
        display();
        break;
    case 4:
        printf("Counter closed. Goodbye!\n");
        return 0;
    default:
        printf("Invalid choice! Try again.\n");
    }
}
}
```

Concept

At a bus ticket counter:

- People **stand in line (FIFO order)** — first person gets served first.
- As new people arrive, they **join the queue at the end**.
- When the counter serves someone, they **leave from the front**.

